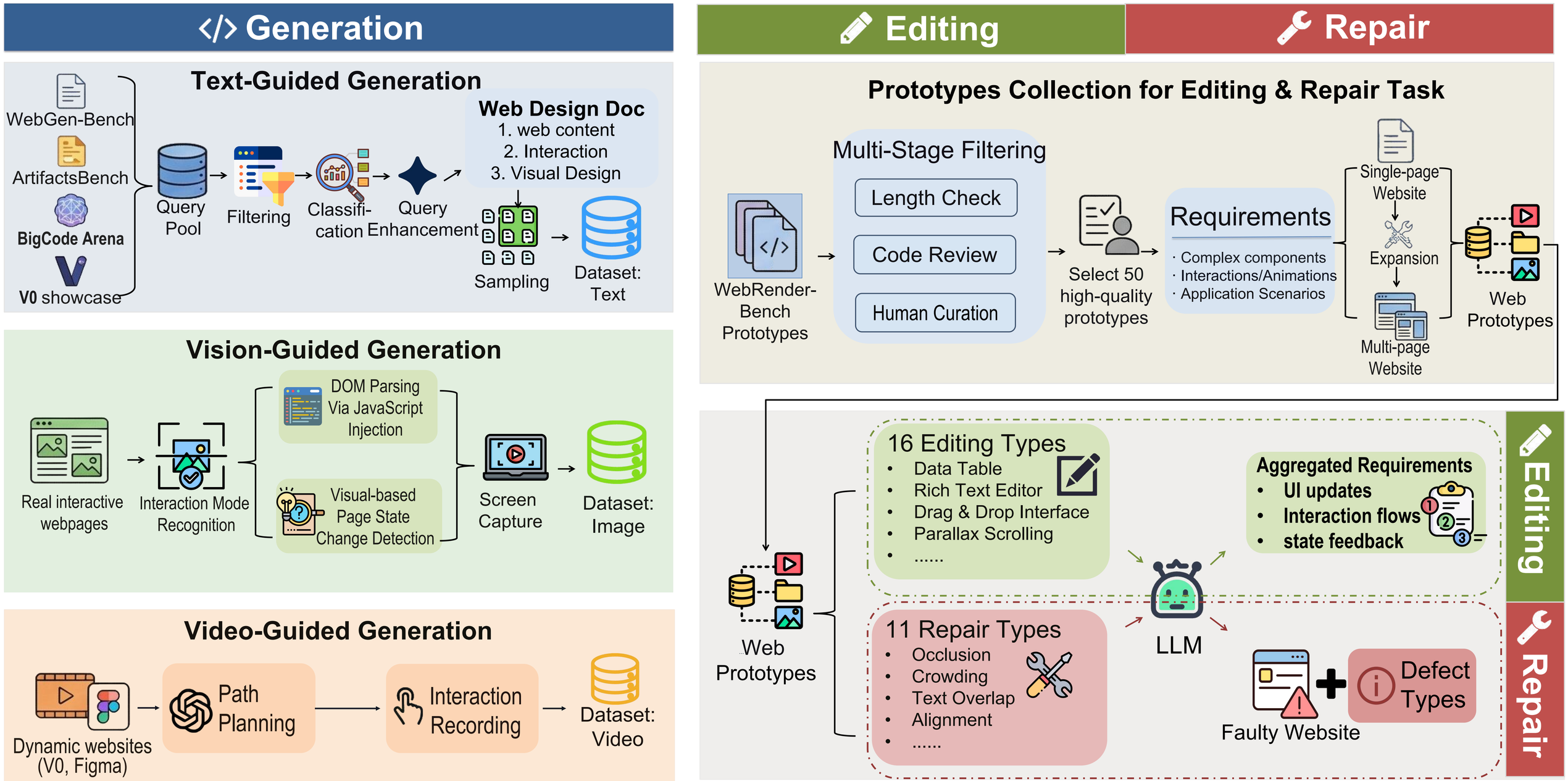
Generation
Editing
Repair
Text-Guided Generation
WebGen-Bench
ArtifactsBench
BigCode Arena
V0 showcase
Query Pool
Filtering
Classifi-cation
Query Enhancement
Web Design Doc
1. web content
2. Interaction
3. Visual Design
Sampling
Dataset: Text
Vision-Guided Generation
Real interactive webpages
Interaction Mode Recognition
DOM Parsing Via JavaScript Injection
Visual-based Page State Change Detection
Screen Capture
Dataset: Image
Video-Guided Generation
Dynamic websites (V0, Figma)
Path Planning
Interaction Recording
Dataset: Video
Prototypes Collection for Editing & Repair Task
WebRender-Bench Prototypes
Multi-Stage Filtering
Length Check
Code Review
Human Curation
Select 50 high-quality prototypes
Requirements
· Complex components
· Interactions/Animations
· Application Scenarios
Single-page Website
Expansion
Multi-page Website
Web Prototypes
Web Prototypes
16 Editing Types
Data Table
Rich Text Editor
Drag & Drop Interface
Parallax Scrolling
.....
Aggregated Requirements
UI updates
Interaction flows
state feedback
LLM
11 Repair Types
Occlusion
Crowding
Text Overlap
Alignment
.....
Faulty Website
Defect Types
Editing
Repair